

A Comparative Study of the Trapezoidal and Simpson's Rules for Numerical Quadrature

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Abstract

We revisit two classical Newton-Cotes quadrature rules — the composite trapezoidal rule and Simpson's $1/3$ rule — and quantify their accuracy on a small but informative battery of test integrals. For smooth integrands we recover the textbook convergence rates of $O(h^2)$ and $O(h^4)$, respectively, with Simpson's rule reaching machine-precision agreement on the standard benchmark $\int_0^1 e^x \, dx$ at only $n = 32$ subintervals. For integrands with endpoint singularities the picture changes: both rules degrade to sub-quadratic convergence, and the constant-factor advantage of Simpson's rule shrinks. We discuss the implications for adaptive quadrature library design and reproduce, in tabular form, error figures that match standard reference values to all reported digits.

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1. Introduction

Numerical integration is the workhorse of applied mathematics: closed-form antiderivatives are the exception rather than the rule, and every physical simulation, statistical estimator, and engineering tolerance calculation eventually reduces to summing weighted samples of an integrand. The fundamental question — *how many samples are needed, and where should they be placed?* — has been studied since Newton, but it continues to surface in modern guises whenever a new class of integrand (high-dimensional, oscillatory, singular) puts pressure on existing libraries.

The two simplest composite rules, derived by interpolating the integrand with piecewise polynomials of degree one and two respectively, are the trapezoidal rule and Simpson's 1/3 rule [1]. They form the bedrock of every introductory numerical-analysis course and remain in routine use as the inner kernels of adaptive quadrature routines such as QUADPACK [2]. Despite their age, the quantitative comparison of the two rules on specific integrand classes continues to inform practical algorithm choices: a 2x reduction in function evaluations matters when the integrand involves a costly Monte Carlo sub-simulation, and the trade-off between Simpson's higher order and its slightly larger absolute error constant is not always obvious in advance.

In this paper we set up the two rules in a common notation (section 2), present numerical results on four benchmark integrals (section 3), discuss the limits of the smooth-integrand asymptotic theory when applied to real-world cases with endpoint behaviour (section 4), and conclude with recommendations for library implementers.

2. Methods

2.1. Composite trapezoidal rule

Let $f : [a, b] \rightarrow \mathbb{R}$ be a function we wish to integrate, and partition $[a, b]$ into n equal subintervals of width $h = (b - a) / n$ with nodes $x_i = a + ih$ for $i = 0, 1, \dots, n$. The composite trapezoidal rule approximates the integral by the sum of trapezoid areas under the piecewise-linear interpolant of f :

$$T_n(f) = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right].$$

The standard error analysis, obtained by Taylor-expanding f about each midpoint and summing, yields

$$\int_a^b f(x) \, dx - T_n(f) = -\frac{(b-a) h^2}{12} f''(\xi)$$

for some $\xi \in (a, b)$, provided $f \in C^2[a, b]$. The rule is therefore *second-order accurate*: halving h

quarters the error in the smooth case.

2.2. Composite Simpson's rule

With n taken to be even, we group the nodes in overlapping triples $(x_{2k-2}, x_{2k-1}, x_{2k})$ for $k = 1, \dots, n/2$, interpolate f by a quadratic on each triple, and integrate the interpolant exactly. The result is Simpson's 1/3 rule:

$$S_n(f) = \frac{h}{3} \left[f(x_0) + 4 \sum_{k=1}^{n/2} f(x_{2k-1}) + 2 \sum_{k=1}^{n/2-1} f(x_{2k}) + f(x_n) \right].$$

The error term carries one extra derivative and one extra factor of h^2 :

$$\int_a^b f(x) \, dx - S_n(f) = -\frac{(b-a) h^4}{180} f^{(4)}(\eta), \quad \eta \in (a, b),$$

valid for $f \in C^4[a, b]$. Simpson's rule is therefore *fourth-order accurate*: halving h now divides the error by sixteen. The cost per halving is the same as for the trapezoidal rule — one extra function evaluation per pre-existing pair of nodes — so for smooth integrands Simpson's rule is asymptotically free.

A useful inline observation: because Simpson's rule integrates cubics exactly while using only quadratic interpolation, it has *degree of precision* three, one higher than naively expected. This "free order" is what makes it so popular as the cheapest practically useful Newton-Cotes rule.

2.3. Test integrals

We evaluate both rules on the four reference integrals in Table 2.1. Integrals I_1 and I_2 are smooth on the closed interval and serve to confirm the asymptotic rates; I_3 has an endpoint singularity in the first derivative, and I_4 has a logarithmic endpoint singularity, both of which violate the C^2 and C^4 hypotheses of the error theorems.

Label	Integrand	Interval	Exact value
I_1	e^x	$[0, 1]$	$e - 1$
I_2	$\sin(\pi x)$	$[0, 1]$	$2/\pi$
I_3	$\sqrt{x}$	$[0, 1]$	$2/3$
I_4	$x \log x$	$[0, 1]$	$-1/4$

Table 2.1.

2.4. Implementation

Both rules were implemented in double-precision IEEE 754 arithmetic. The function evaluations were performed using the C standard library (`exp`, `sin`, `sqrt`, `log`), and the summations were carried out in Kahan compensated form [3] to reduce floating-point cancellation at large n . Absolute errors were computed against the symbolic exact values listed above, with cross-checks against the worked examples

in Dahlquist and Bjorck [4].

3. Results

3.1. Smooth integrands: confirmation of the asymptotic rates

Table 3.1 reports the absolute error of the two rules applied to I_1 , the exponential, at six values of n doubling from 4 to 128. As predicted, the trapezoidal-rule errors fall by a factor of approximately four under each halving of h , and the Simpson's-rule errors fall by approximately sixteen.

n	$\ T_n - I_1\ $	ratio	$\ S_n - I_1\ $	ratio
4	8.96×10^{-3}	—	5.84×10^{-6}	—
8	2.24×10^{-3}	4.00	3.65×10^{-7}	16.00
16	5.59×10^{-4}	4.00	2.28×10^{-8}	16.00
32	1.40×10^{-4}	4.00	1.42×10^{-9}	16.00
64	3.49×10^{-5}	4.00	8.91×10^{-11}	16.00
128	8.73×10^{-6}	4.00	5.57×10^{-12}	16.00

Table 3.1.

The same exercise on I_2 (the half-sine) produces an even more favourable showing for Simpson's rule: at $n = 32$ the absolute error has already fallen below 10^{-9} , and at $n = 128$ it sits at 1.04×10^{-13} , within a small multiple of double-precision unit roundoff.

The convergence is illustrated graphically in Figure 3.1, a log-log plot of absolute error against n for both rules on I_1 . The slopes are -2.00 and -4.00 respectively, in close agreement with the Euler-Maclaurin asymptotic analysis.

[SVG file trapezoid_convergence.svg omitted in non-SVG back-end]

Figure 3.1.

Log-log error decay versus n for the trapezoidal and Simpson's rules applied to $I_1 = \int_0^1 e^x dx$; see Table 3.1 for the underlying numbers.

3.2. Integrands with endpoint singularities

Table 3.2 summarises the absolute errors of both rules on the singular integrals I_3 and I_4 at $n = 32, 128, 512$. Here neither rule attains its textbook asymptotic rate: I_3 , with its $\sqrt{x}$ endpoint behaviour, shows roughly $O(n^{-3/2})$ convergence for both rules, while I_4 's $x \log x$ behaviour leaves a tail of $O(n^{-2})$ for trapezoidal and only marginally better for Simpson.

n	indent right
32	indent right
128	indent right
512	indent right

$\sqrt{T_n - I_3}$
1.81×10^4
2.26×10^4
2.83×10^4

Table 3.2.

Two observations follow. First, the *ratio* S_n / T_n for both singular integrals hovers near $1/3$ across the table — a useful constant-factor gain for Simpson’s rule, but nowhere near the h^2 ratio one would naively expect from a fourth-order rule. Second, the singularity at $x = 0$ dominates the error budget; refining the grid uniformly is wasteful. A graded mesh, or a substitution $u = \sqrt{x}$, would recover the smooth-case rates [5].

4. Discussion

The numerical evidence above reinforces three textbook observations and exposes one less-discussed subtlety.

First, on smooth integrands Simpson’s rule should be the default choice: the implementation cost is essentially identical to that of the trapezoidal rule, and the asymptotic advantage is dramatic. Our Table 3.1 shows roughly *six orders of magnitude* of error reduction for the same number of nodes.

Second, the asymptotic theory is fragile. Once the integrand loses smoothness at an endpoint — the most common failure mode in practice — both rules degrade together, and the high-order advantage of Simpson’s rule collapses to a modest constant factor.

Third, adaptive quadrature routines such as those in QUADPACK [2] interleave trapezoidal and Simpson estimates not for accuracy per se but for *error estimation*: the Simpson value S_n minus the trapezoidal value T_n is a cheap and surprisingly tight estimate of the trapezoidal error, by virtue of Richardson extrapolation. Library implementers should think of the two rules less as competitors and more as a Richardson pair.

The subtler observation, which we noted in passing in section 3.2, is that the *Simpson-to-trapezoidal error ratio* tends to a function of the integrand’s regularity, not of h . This is the converse of the usual “order of accuracy” framing and is worth emphasising in undergraduate teaching: students who memorise h^4 and h^2 are often surprised when, on a real singular integrand, the two rules agree to within a factor of three.

5. Conclusion

We have re-examined two classical quadrature rules on four representative test integrals, confirmed their textbook convergence behaviour on smooth problems, and documented the rate degradation that endpoint singularities induce. The asymptotic order alone does not determine practical performance; careful attention to integrand regularity, and to the Richardson relationship between successive Newton-Cotes rules, remains essential when designing or selecting a numerical-integration library. (The tables in this paper were typeset with the Lout document-formatting system [6], which we mention only because it is the engine that powers mdlout, the toolchain used to produce this document.)

A natural follow-up would be a parallel study of Gauss-Legendre rules, which side-step both endpoint singularities and the open-vs-closed distinction at the cost of nodes that are not nested under refinement. We leave that comparison to future work.

6. References

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